ESP32 WiFi

Safety Scanner

OLED Display + Multi-Network Analysis

Educational IoT Project

Engineer Amel

@engineeramel

📡 WiFi Scan

📊 3 OLED Views

🔐 Security Check

⚡ Auto Loop

WORKING SUMMARY

How It Works

1
Initialize
WiFi & OLED ready (STA Mode + SSD1306 128×64)

2
Scan
WiFi.scanNetworks() Collects SSID, RSSI, CH, ENC

3
Process
Calculates Quality %, Speed, Risk %, Safety Level

4
Display
3 views per network: Identity → Quality → Security

5
Loop
Auto cycles all networks then rescans automatically

ESP32 → scans → processes → displays (3 views per WiFi) → repeats

OLED DISPLAY

3 Views Per Network

Each view stays ~5 seconds before auto-switching

View 1 — Identity

```
SSID: HomeNet

RSSI: -65 dBm

CH: 6

SEC: WPA2
```

Who is this network?

View 2 — Quality

```
SSID: HomeNet

QUAL: 78 %

SPD: 54 Mbps

ENC: HIGH
```

How good is the signal?

View 3 — Security

```
SSID: HomeNet

ENC: WPA2

RISK: 35 %

SAFE: MEDIUM
```

How safe is this network?

Display Symbols Explained

Symbol	Meaning	Example on OLED
SSID	WiFi network name	`SSID: HomeNet`
RSSI	Signal strength (dBm)	`RSSI: -65 dBm`
CH	Channel number	`CH: 6`
SEC	Security type	`SEC: WPA2`
QUAL	Signal quality (%)	`QUAL: 78 %`
SPD	Estimated speed (Mbps)	`SPD: 54 Mbps`
ENC	Encryption strength/type	`ENC: HIGH`
RISK	Hack chance % (educational)	`RISK: 35 %`
SAFE	Safety level text	`SAFE: MEDIUM`

View Groups

View 1

SSID, RSSI, CH, SEC

View 2

SSID, QUAL, SPD, ENC

View 3

SSID, ENC, RISK, SAFE

Accuracy Truth Table

Feature	Accuracy	Limitation
RSSI (Signal Strength)	High (≈98%)	Small fluctuation (±2 dBm)
Channel Detection	100%	None
Encryption Type	100%	None
Signal Quality %	Medium-High (≈90%)	Derived, not hardware measured
Estimated Speed (Mbps)	Low-Medium (≈50-60%)	Only an approximation
Hack Chance %	Conceptual	Not real hacking, just awareness
Display (OLED)	Clear but small	Limited space for details
Power Usage	Moderate	Drains battery on long scans

Strong & Reliable
RSSI, Channel, Encryption

Approximate
Quality %, Estimated Speed

Educational Only
Hack Chance %

Physical Limit
OLED size, Power

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ADVANCED FEATURES

Advanced Functions & Benefits

Auto Multi-Network Scan

Scans 2+ WiFi signals continuously

Full coverage

3-View OLED Display

Identity → Quality → Security

Clear info, step-by-step

Signal Quality Estimation

Converts raw RSSI to % value

Easy to understand strength

Security Awareness

Shows encryption type & safety level

Informs about risks

Smart Loop Function

Auto cycles & rescans networks

Hands-free operation

Efficiency Optimized

No lag / smooth display updates

Reliable performance

PRO vs CON

Advantages vs Disadvantages

Aspect	✅ Advantage	❌ Disadvantage
Cost	✅ Very cheap hardware	❌ Limited compared to pro tools
Portability	✅ Small, handheld device	❌ Small screen (128×64)
Functionality	✅ Scans SSIDs, RSSI, security	❌ No deep packet analysis
Accuracy	✅ Good for hobby use	❌ Not 100% precise
Use Case	✅ Education, diagnostics	❌ Misuse → illegal consequences

LEGAL WARNING

WiFi Scanner — Use Cases

Use Case	Allowed?	Notes
Educational (learning, school, lab)	✅ Yes	Safe usage
Signal strength & coverage check	✅ Yes	Helps optimize WiFi
IoT device setup (list SSIDs)	✅ Yes	Common safe use
Security awareness (OPEN vs WPA2)	✅ Yes	Educational only
Cracking passwords / hacking WiFi	❌ No	Illegal
Attacking or jamming WiFi	❌ No	Illegal
Monitoring private networks (no consent)	❌ No	Privacy violation

⚠️ **Use only for scanning, learning & testing your OWN WiFi. Do NOT use for hacking or disrupting others — it is illegal!**

Engineer Amel

@engineeramel

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